

Four Types of Cyclists?

Examination of Typology for Better Understanding of Bicycling Behavior and Potential

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The labeling and the categorizing of cyclists have been occurring for more than a century for a variety of purposes. This study examined a typology developed by the City of Portland, Oregon, that included four categories of cyclists: “the strong and the fearless,” “the enthused and confident,” “the interested but concerned,” and “no way, no how.” Unlike several other typologies, this widely referenced typology was intended to apply to all adults, regardless of their current cycling behavior. An analysis was conducted with data from a random phone survey ($n = 908$) of adults in the Portland region. Adults were placed into one of the four categories primarily on the basis of their stated comfort level with cycling on a variety of facility types, their interest in cycling as transportation, and their physical ability to bicycle. Nearly all of the sampled population fit clearly into one of the four categories. A majority (56%) of the region’s population fit in the interested but concerned category, which was thought to be the key target market to increase the use of cycling as a form of transportation. The analysis indicated that a reduction in traffic speed and an increase in the separation between bicycles and motor vehicles (e.g., cycle tracks) might increase levels of comfort and cycling rates. Women and older adults were underrepresented among the more confident adults and those who currently cycled for transportation.

Typologies and classifications arise out of a desire to understand populations and apply categories that further a goal, be it academic or professional. From the dawn of bicycling in the United States, riders have been classified. To accommodate the burgeoning interest in bicycles and their predecessors, velocipedes, special bicycle riding rinks were created. Managers of these rinks created one of the first cyclist classification systems, which isolated “timid toddlers,” who were still learning to ride these machines, from the classes of successively skilled riders in the rinks: the “wary wobblers,” “go-it-gracefuls,” and the “fancy few” (1).

A 1994 FHWA report sought to outline bicycle facility design concepts that would cater to all cyclist types, which, the report proposed, included “advanced bicyclists” who could “operate under most traffic conditions”; “basic bicyclists,” who were “less confident of their ability to operate in traffic without special provisions for bicycles”; and “children,” who bicycled under parental supervision (2). The 1999 AASHTO *Guide for the Development of Bicycle Facilities* echoed the FHWA report’s A, B, C of bicycle user groups, and stated

that “these three bicycle user types are a helpful guide to the highway designer” (3). These references made only vague efforts to quantify the share of people in each group. The FHWA report suggested that perhaps only 5% of the people that owned bicycles could be classified in the advanced category (2), while the AASHTO guide simply stated that “some” adults fell into the advanced category, but “most” fell into the basic category (3). These typologies categorized existing bicycle users on the basis of their skill level but did not seek to categorize cyclists on the basis of their purpose (e.g., recreation, transportation), nor did they explicitly consider those that were not currently bicycle users.

BACKGROUND AND LITERATURE REVIEW

In 2006, the Bicycle Coordinator for the City of Portland, Oregon, Roger Geller, released a paper that proposed a new typology of cyclists, entitled “Four Types of Cyclists” (4). The paper focused on riding for transportation purposes, and suggested that cyclists fell into one of four categories: “the strong and the fearless,” “the enthused and confident,” “the interested but concerned,” or “no way, no how.” Geller’s categories were in part determined by a person’s comfort on a bicycle on different types of bikeways (4). Strong and fearless cyclists ride “regardless of roadway conditions.” Enthused and confident cyclists are comfortable to ride on a road with automobiles, but “prefer to do so operating on their own facilities” and appreciate efforts made to improve the bikeway infrastructure. Interested but concerned people are “curious about bicycling,” like to ride but are afraid to do so, and therefore do not regularly ride and “will not venture out onto the arterials.” Finally, the no way, no how group is not going to ride a bicycle, “for reasons of topography, inability, or simply a complete and utter lack of interest.” This categorization was intended to cover all adults, regardless of their current bicycling behavior. The paper suggested that the strong and the fearless made up less than 1% of the city of Portland’s population, while the enthused and confident constituted perhaps 7%. The interested but concerned were posited to comprise about 60% of the city. The no way, no how people comprised the remainder of the population, or about 33%. In its new Bicycle Master Plan, the city targeted the interested but concerned group as the market necessary to reach to achieve ambitious transportation mode shift targets. As a result, the plan emphasized “low-stress” facilities, such as bicycle boulevards and separated cycle tracks.

Geller’s paper has generated much discussion among bicycle bloggers and advocates on well-read websites such as Bikeportland.org (4, 5), Planetizen (6), and Reconnecting America (7). It has also gained considerable traction with bicycle planners. The study reported here identified at least 14 recent city or regional bike plans

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(e.g., final, draft, supporting documents) that referenced Geller's typology (4), either with or without attribution: Cambridge, Ontario, Canada [2008 (8)]; Albany, New York [2009 (9)]; Burlington, Ontario, Canada [2010 (10)]; Palo Alto, California [2011 (11)]; Los Angeles, California [2011 (12)]; South Bay, California [2011 (13)]; Sunshine Coast, Queensland, Australia [2011 (14)]; Reno–Sparks, Nevada [2011 (15)]; Lincoln City, Oregon [2011 (16)]; Melbourne, Victoria, Australia [2012 (17)]; Southern California Association of Governments, Los Angeles [2012 (18)]; Bloomington–Normal, Illinois [2012 (19)]; Lower Savannah Council of Governments, Georgia [2012 (20)]; and Seattle, Washington [2012 (21)].

Many of the plans used the typology to demonstrate why investments in bicycle facilities were worthwhile. The 2011 plan for Reno–Sparks, Nevada, noted that Geller's typology showed how “a potential expansion of bicyclists could be attracted by investing in a better, safer bikeway system” (15, p. 27). The 2011 plan for Palo Alto argued that the city should plan for the interested but concerned group, and directly quoted Geller's paper to state that “riding a bicycle should not require bravery” (11, p. 5-13). Others simply used the typology to support the idea to build bikeways that were comfortable for a wide range of people. The plan for Cambridge, Ontario, Canada, stated that the four types of cyclists “illustrate that there is great potential to change the behavior of a large proportion of the population, if changes to transportation infrastructure address the perception of cycling safety and comfort” (8, p. i). The plan for Sunshine Coast, Queensland, Australia, stated that “one of the main goals of an active transport plan is to convert noncyclists to ‘enthused and confident’ cyclists” (14, p. 38). Although the percentile breakdowns of the four types of cyclists were qualified in Geller's paper as reasoned estimates, many of the citations of the typology did not include these qualifications. Sometimes the plans and studies used Geller's typology concept (4), but adapted the categories to fit their goals or perception of cyclist categories. For example, instead of no way no how, Calgary called the group least likely to cycle the “reluctant to cycle” (22, p. 2). Seattle used the term “willing but wary” in place of interested but concerned (23, p. 2). Note that the recent draft Seattle Bike Plan used the interested but concerned phrasing (21, p. 3).

In the academic realm, some researchers have categorized cyclists to better understand the nonhomogeneous behavior among people that bicycle. Several studies use current cycling frequency to categorize cyclists. Winters et al. defined everyone that had not ridden a bicycle in the past year as a “potential cyclist,” while all others either were “occasional,” “frequent,” or “regular” cyclists (24). Similarly, Heinen et al. classified commuters into one of three groups [e.g., “noncyclists,” “full-time cyclists” (every working day), “part-time” cyclists (at least once a year)], and identified differences in attitudes between these groups (25). In their analysis of workers in two Swedish cities, Bergstrom and Magnusson added a seasonal dimension to their categories: “winter cyclist,” “summer-only cyclist,” “infrequent cyclist” (less than 2 of 5 days a week), and a “never” cyclist (26). They identified the relative importance of factors that influenced the decision to commute by bicycle, such as exercise, cost, and the environment, and showed how they differed by category. Gatersleben and Haddad used factor analysis to identify four types (or stereotypes) of cyclists: “responsible,” “lifestyle,” “commuter,” and “hippy-go lucky” on the basis of the answers they received to 50 questions about the attributes of cyclists (27). Such categories and their discussion can provide a useful framework to understand how people view cyclists. None of these studies attempted to estimate the share of the broader population that fit into each category.

In a study for London's Department for Transportation, Christmas et al. discussed the difficulty involved in the attempt to segment the cycling population. They concluded that the method must depend upon the intended purpose (28). For their purpose of road safety, the authors suggested including all or some of five variables: age, gender, motivation for cycling, cycling patterns, and cycling approaches. Although they did not develop a typology, they noted significant diversity within the population cycling for utility (versus for leisure) and the likelihood that individuals might belong to more than one group.

OBJECTIVES

Geller's four types of cyclists have clearly resonated among many transportation professionals, and they are now helping to guide bicycle planning in a growing number of cities (4). Given the expansive role of Geller's typology, it is useful to examine its origins and validity. Geller described the intent and process used to develop his typology as follows:

The intent behind its development was to get a better handle on our market for bicycle transportation. . . . As stated previously, the numbers assigned to each of these categories are not something over which any bicycle planner should be prepared to fall on their sword. These numbers, when originally assigned, were not based upon any survey or polling data, or on any study. Rather, they were developed based on the professional experience of one bicycle planner. Soon thereafter, these numbers were discussed and, in effect, vetted with various informed individuals and groups. . . . But beyond that initial vetting of the idea there has been survey, polling and study data that continues to support the assignment of both numbers to and description of the categories. (4)

The objectives of the study reported in this paper were to (a) examine the validity of Geller's four types of cyclists in the Portland, Oregon, region (4), (b) understand who fell into each type, and (c) use the typology to explore what actions might increase levels of cycling for transportation.

METHODOLOGY

The data were obtained through a random phone survey of adults in the Portland region. The sample included both landline and mobile phone numbers and was conducted July 19 through August 10, 2011. Nine hundred and two interviews were completed. Of those, 130 (14%) were completed on mobile phones. The mobile phone sample was used to help reduce sampling bias, in particular among younger adults (29). The overall response rate was 20% of eligible numbers and 35% of resolved numbers [see American Association for Public Opinion Research (30) for definitions]. The data were weighted by age and sex to reflect the population of the region with the use of the 2010 U.S. Census. Respondents that indicated that they were physically unable to ride a bicycle answered a subset of the questions, which focused on attitudes and demographics. Of the weighted sample, 13% fell into this category, including about 40% of the respondents that were 65 or older.

The assignment of the sample into the four categories required a careful examination of the typology. A few things distinguish Geller's typology from the others described earlier (4). First of all, the primary intent is to understand the market for cycling, not just the population of current cyclists. The typology does not have its basis solely in the current riding behavior. Second, the focus is on

cycling done for transportation and not for recreation or leisure. A close examination of Geller's descriptions reveals that his typology has its basis primarily in people's comfort level (e.g., fearless, confident, concerned) to ride on different types of facilities and secondly in people's interest or intent to bicycle [e.g., enthused, interested, no way (4)]. Actual bicycling behavior is not a primary factor in the determination into which category someone falls.

The first step of the categorization process in the study reported here was to use the survey sample, which had its basis in a series of questions about the comfort level with cycling on various types of streets. For each hypothetical scenario, the respondents were asked to indicate their level of comfort on a scale of 1 to 4: 1 meant "very uncomfortable" and 4 meant "very comfortable." The scenarios used for the categorization were as follows:

1. A path or trail separate from the street;
2. A two-lane, neighborhood, commercial-shopping street with traffic speeds of 25 to 30 mi per hour, on-street car parking, and no bike lane: What if a striped bike lane was added?
3. A major urban or suburban street with four lanes, on-street parking, traffic speeds of 30 to 35 mi per hour, and no bike lane:
 - What if a striped bike lane was added?
 - What if it also had a wide bike lane separated from traffic by a raised curb or parked cars? and
4. A major street with two lanes in each direction, a center divider, on-street parking, traffic speeds of 35 to 40 mi per hour, and no bike lane:
 - What if a striped bike lane was added?
 - What if it also had a wide bike lane separated from traffic by a raised curb or parked cars?

Level of comfort was determined primarily by the responses to the three scenarios that involved streets with and without bike lanes. These scenarios best matched Geller's descriptions. Geller described the strong and the fearless as people that were willing to ride regardless of roadway conditions. Therefore, this group was defined as very comfortable on nonresidential streets without bike lanes. An average of 3.5 points or higher in response to those three questions was defined as "very comfortable." To meet that description, a respondent would need to have answered "4" for comfort level in at least two of the scenarios and "3" in the other. Geller described the enthused and confident as people that were comfortable sharing the roadway with

cars but preferred to do so with their own facilities, such as bike lanes. Respondents that had an average comfort level of 3.5 or higher for the three nonresidential street scenarios with bike lanes were put into this category. They could be considered very comfortable on nonresidential streets with bike lanes. At the other end of the scale, respondents that indicated that they were very uncomfortable riding a bicycle on a path or trail separate from the street were put into the no way, no how category, along with the respondents that were physically unable to ride a bicycle.

This first step left a share of the respondents uncategorized. They were respondents that did not feel "very comfortable" on nonresidential streets, with or without bike lanes, but did not feel "very uncomfortable" on paths and trails. The second step used respondents' interest in cycling to determine whether these respondents should be either interested but concerned or no way, no how. Respondents were asked to agree or disagree (strongly or somewhat) to the statement "I would like to travel by bike more than I do now." Those that agreed with this statement were put in the interested but concerned category, and those that disagreed were put in the no way, no how category. However, an examination of actual cycling behavior revealed that some people that cycled for transportation in the past 30 days ended up in the no way no how category. This finding made sense, because some people that currently cycled might have had no interest to cycle more. As a final step, respondents were put into the interested but concerned category, who were not very comfortable to cycle on nonresidential streets, were not interested in cycling more, but who had cycled for transportation in the past 30 days. The need to place them in this category pointed to the difficulty involved in the categorization of people on the basis of multiple dimensions; in this case, on the basis of comfort level and interest. Nearly all (91%) of the adults placed in the enthused and confident category were interested in cycling more, which made the "enthused" part of the label largely accurate.

FINDINGS

Distribution of Respondents by Cyclist Type

The distribution of survey respondents into the four types appears in Table 1. The distribution is similar to Geller's estimate (4), although with a higher share of the adult population in the strong and the

TABLE 1 Distribution of Survey Respondents by Cyclist Type

Type	Description	City of Portland ^a (%)	Rest of Region ^b (%)	All ^c (%)	Geller's Estimate for City ^d (%)
Strong and the fearless	Very comfortable without bike lanes	6	2	4	<1
Enthused and confident	Very comfortable with bike lanes	9	9	9	7
Interested but concerned	Not very comfortable, interested in biking more Not very comfortable, currently cycling for transportation but not interested in biking more	60	53	56	60
No way, no how	Physically unable Very uncomfortable on paths Not very comfortable, not interested, not currently cycling for transportation	25	37	31	33

NOTE: Weighted data may not total 100% as a result of rounding.

^a*n* (weighted) = 436.

^b*n* (weighted) = 479.

^c*n* (weighted) = 915.

^dGeller, 2012(4).

TABLE 2 General Cycling Behavior by Cyclist Type

Type	Behavior (%)				Total [% (number of respondents)]
	Utilitarian	Recreational	Noncyclist	Unable or Don't Know	
Strong and the fearless	43	23	34	na	100 (35)
Enthusied and confident	46	31	23	na	100 (82)
Interested but concerned	43	30	28	na	100 (511)
No way, no how	na	15	46	40	100 (287)

NOTE: Weighted data may not total 100% as a result of rounding. na = not applicable.

fearless and the enthused and confident categories and a smaller share in the no way no how category.

Respondents were then categorized on the basis of their current cycling behavior into one of the following three groups:

- Utilitarian cyclists. Cyclists who had cycled at least once in the past 30 days for transportation (e.g., for work, school, shopping) and usually cycled once a month for transportation in a typical summer or winter month.
- Recreational cyclists. Cyclists who had cycled at least once in the past 30 days but did not meet the threshold for utilitarian cyclists.
- Noncyclists. Cyclists who had not cycled in the past 30 days or individuals who stated that they “never ride a bicycle” (a screening question).

Given these definitions, a recreational cyclist was someone that cycled at least once for transportation in the past month but did not

cycle at least once a month in a typical summer or winter month for transportation. The utilitarian cyclist category was for people that had developed some pattern of cycling for transportation that extended beyond the past month.

Contrary to what might have been expected, similar shares of three of the four types (not including no way, no how) could be considered utilitarian cyclists (Table 2); 43% of the strong and the fearless, 46% of the enthused and confident, and 43% of the interested but concerned were classified as utilitarian cyclists.

Although the distribution of the types within these broader cycling behavior categories was similar, the actual amount of cycling did differ significantly between the types. Over a given month, the enthused and confident respondents cycled an average of 9.7 days, of which they cycled 4.2 days for work or school. This rate was significantly higher than that of the interested but concerned (6.2 days overall, 1.6 day for work or school) and no way, no how (2.4 days, all for recreation). Figure 1 shows the number of days that utilitarian

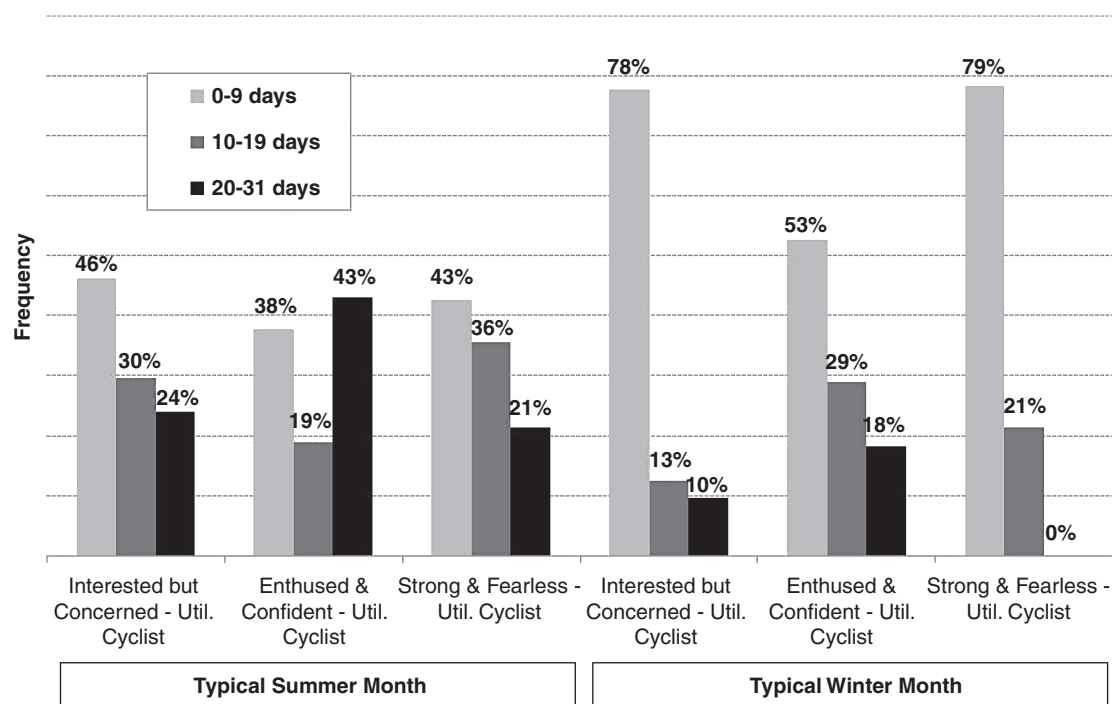


FIGURE 1 Frequency of utilitarian cyclists' cycling for transportation by cyclist type (util. = utilitarian).

cyclists typically rode for transportation in both summer and winter months. The most significant differences occurred between the interested but concerned and the enthused and confident groups; 24% of the former, compared with 43% of the latter, cycled 20 or more days for transportation in a typical summer month. Although all of the types cycled less often in winter months, the enthused and confident remained the group that cycled most often.

Demographics of Cyclist Types

Many of the bicycle plans noted earlier acknowledged that, if overall rates of cycling were to increase significantly, cycling rates needed to increase among demographic groups that currently did not cycle very much. In most major cities in the United States, men have been found to make up a disproportionate share of utilitarian cyclists (31). In the Portland sample surveyed, women were more likely to be in the no way, no how and interested but concerned categories. Within each of the four types, women were more likely to be noncyclists (Table 3). For example, only 22% of the enthused and confident utilitarian cyclists were women, compared with 47% of the enthused and confident noncyclists.

Older adults in the United States were far less likely to cycle than their counterparts in many European cities (32). This finding surfaced in the data reported here as well. Adults that were 55 years of age and older were overrepresented in the no way, no how category (Table 3). Members of the enthused and confident group that currently cycled tended to be more middle-aged: 68% of them that were utilitarian cyclists were 35 to 54 years old. In contrast, 37% of the interested but concerned utilitarian cyclists were 35 to 54 years old, while 47% were younger than 35. No significant differences were found between the groups with respect to education or income levels.

Whether someone cycled to school as a child appeared to correlate with the category into which they fell. Those in the enthused and confident group were those most likely to have cycled to school when they were children (Table 3). A large majority, more than 60%, of the no way, no how group had never cycled to school as children, an indication that this lack of experience might have influenced their

level of comfort and interest in cycling as adults. However, the pattern was inconsistent. Within the interested but concerned group, the noncyclists actually were the least likely to have never cycled to school.

Understanding of the Market

Attitudes and Perceptions of the Four Types

The intent of the Geller typology was to guide efforts to increase bicycling for transportation. Exploration of the differences between the groups with respect to attitudes and perceptions about cycling may confirm whether the groups are distinct and, if so, explain the differences in cycling behavior described earlier in this paper. Table 4 shows some questions in cases in which there were significant differences between the four groups. In some cases, the no way no how group was the most distinct, with insignificant differences between the other three groups. For example, the no way, no how group was much less likely to agree that they liked to ride a bike, wanted to travel by bike more often, and saw people like themselves on bikes on city streets. The greatest differences among all four groups surfaced in the answers to the questions on traffic and safety. That traffic and safety concerns were the source of the differences made sense, given that the categories were created largely on the basis of comfort levels with cycling on different types of roads. Several significant differences also arose among the four groups as to how easy it was thought to be to ride a bike. The interested but concerned respondents were significantly less likely than the enthused and confident to agree that “biking can sometimes be easier for me than driving” and that “for me to ride a bike for daily travel from home would be easy.”

The City of Portland, as well as many of the jurisdictions that have adopted the Geller typology (4), decided to focus on the interested but concerned group of cyclists as their target market to expand cycling levels. This group was the largest in the study. Moreover, the data showed that in many respects this group was distinct from the no way, no how and the enthused and confident groups. In particular, the level of interest in cycling appeared to be as high as

TABLE 3 Demographics of Cyclist Types

Type	Women (%)	Age (%)			Never Cycled to School as a Child (%)
		18–34	35–54	≥55	
Strong and the fearless	21	86	9	6	50
Enthused and confident: utilitarian cyclist	22	19	68	14	35
Enthused and confident: recreational cyclist	20	20	60	20	36
Enthused and confident: noncyclist	47	32	32	37	37
Interested but concerned: utilitarian cyclist	43	47	37	17	53
Interested but concerned: recreational cyclist	43	34	47	20	51
Interested but concerned: noncyclist	58	22	41	37	42
No way, no how: recreational cyclist	52	29	52	19	68
No way, no how: noncyclist	63	25	33	42	63
No way, no how: unable–don’t know	78	11	17	73	NA
All	51	32	37	31	51

NOTE: The strong and the fearless category is not broken down by current cycling behavior because of the small sample in that group. NA = not available.

TABLE 4 Attitudes and Perceptions by Type

Attitude or Perception	No Way, No How ^a	Interested but Concerned ^b	Enthusied and Confident ^c	Strong and the Fearless ^d
I like riding a bike.	1.9	3.3	3.5	3.1
I would like to travel by bike more than I do now.	1.8	3.4	3.5	3.1
In general, I see people similar to me bicycling on city streets.	2.6	3.0	3.1	3.2
Biking can sometimes be easier for me than driving.	1.4	2.4	2.9	2.8
For me to ride a bike for daily travel from home would be easy.	1.5	2.3	3.1	2.8
Traveling by car is safer overall than riding a bike.	3.5	3.0	2.6	1.9
There is so much traffic along the street I live on that it would make it difficult or unpleasant to bike.	2.0	1.8	1.4	1.3
There is so much traffic along nearby streets that it would make it difficult or unpleasant to bike.	2.7	2.4	1.7	1.6
There are bike lanes that are easy to get to in my neighborhood.	2.5	2.9	3.1	2.6
There are off-street bike trails or paved paths in or near my neighborhood that are easy to get to.	2.5	2.8	2.9	3.1

NOTE: Significant differences exist between the groups for all factors on the basis of a one-way analysis of variance test, $p < .05$. Average level of agreement: 1 = strongly disagree, 4 = strongly agree.

^a $n = 170$ – 173 .

^b $n = 507$ – 511 .

^c $n = 81$ – 82 .

^d $n = 31$ – 34 .

that of the enthused and confident. The differences were that the interested but concerned group had greater concerns about safety, traffic, and cycling ease, and currently cycled less than the enthused and confident group. The final section of this paper focuses on the interested but concerned survey respondents.

Perception of Facility Types: Interested but Concerned

Cities have a range of infrastructure and programmatic tools available to try to increase the share of people that ride a bicycle for transportation. The provision of infrastructure that supports biking (in particular, bike lanes and paths) has been a common approach toward this goal for more than 20 years. More recently, U.S. cities have been experimenting with other types of facilities, including bicycle boulevards and cycle tracks. Bicycle boulevards use traffic calming tools, such as speed humps and traffic diverters, on low-volume streets, to slow down and reduce the volume of motor vehicle traffic. Coupled with signage, these facilities have been shown to attract cyclists (33). Cycle tracks operate similarly to bike lanes along major streets but incorporate a physical barrier between motor vehicles and bicycles (e.g., curb, bollards, parked cars). The survey's questions about comfort level on various types of streets included these types of facilities and briefly described them.

The survey responses revealed some significant differences in comfort level with a regular striped bike lane and a cycle track (Figure 2). Comfort levels were highest on a separated path or trail, followed closely by quiet residential streets, with or without traffic calming features. On nonresidential streets, comfort levels rose in general as separation increased between cyclists and motor vehicles. A cycle track facility on a four-lane street without a center divider and speeds of 30 to 35 mph yielded a level of comfort (3.2)

that approached that of a quiet residential street (3.6) or separate path (3.7).

Perceptions of Physical Environment Factors

Within the interested but concerned group, slightly more than 40% currently cycled to some extent for transportation. An examination of the differences between this subgroup and those that currently did not cycle, or that cycled primarily for recreation, might reveal barriers to cycling that could be addressed through infrastructure or programs. Table 5 shows the share within each of the interested but concerned subgroups that agreed (strongly or somewhat) with a series of statements about their neighborhood. The noncyclists were significantly less likely than the utilitarian cyclists to agree that bike lanes were available and easy to access and more likely to agree that the high volume of traffic on nearby streets would make it difficult or unpleasant to bike.

Perceptions of Social and Personal Factors

Several survey questions were intended to address social and personal factors that might affect decisions to cycle (Table 6). Within the interested but concerned group, levels of social support and influence appeared to be significantly lower for the noncyclists and recreational cyclists, compared with the utilitarian cyclists. For example, only 17% of noncyclists lived with people that bicycled for transportation, compared with 53% of the utilitarian cyclists. Fewer of the noncyclists indicated that they had coworkers who biked to work or that they saw people similar to themselves on bikes on city streets.

The noncyclists and recreational cyclists also seemed to encounter more personal barriers that prevented them from cycling. They

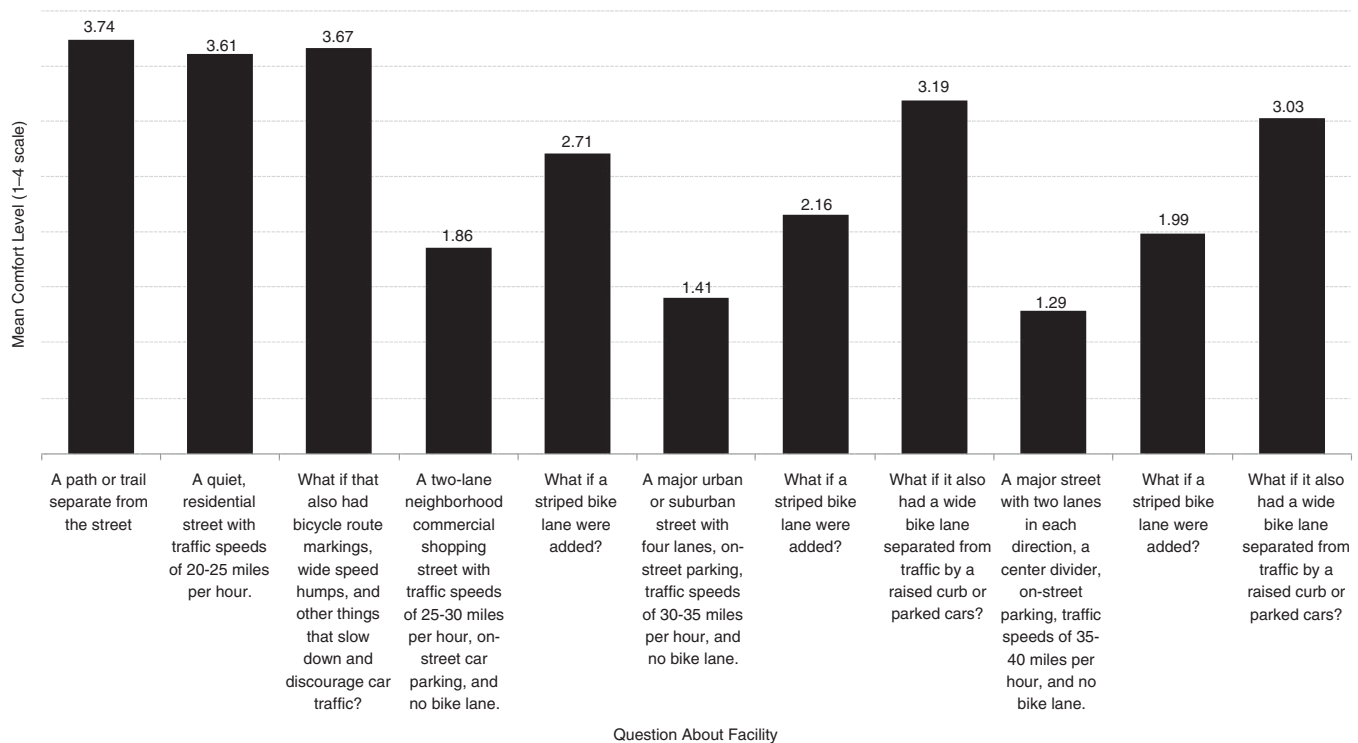


FIGURE 2 Average level of comfort cycling on various facilities: interested but concerned group.

were less likely to agree that bicycling for daily travel would be easy, or that their destinations were within biking distance. They were more likely to say that they did not have time to bike instead of drive. These barriers were related to the physical environment, with respect to infrastructure and land use. Clothing and helmets might not have been a significant barrier: 72% of the noncyclists indicated that to commute by bike would require them to wear different clothing, which was not significantly higher than the 69%

of utilitarian cyclists that agreed. Less than one-third of the noncyclists indicated that they did not like to wear a helmet, a response that was similar to that of the other subgroups. The large majority (83%) of the noncyclists indicated that they knew how to ride safely in traffic, although 63% indicated a desire to learn how to ride more safely in traffic. The noncyclists also were less likely to feel comfortable to ride in the rain and after dark. The three subgroups were equally and highly (more than 80%) likely to be

TABLE 5 Perceptions of Physical Environment Among Interested but Concerned Group

Perception of Physical Environment	Interested but Concerned (% agreeing)		
	Noncyclist ^a	Recreational Cyclist ^b	Utilitarian Cyclist ^c
There are off-street bike trails or paved paths in or near my neighborhood that are easy to get to.	63	64	70
There are bike lanes that are easy to get to.	66	62	78
There are quiet streets, without bike lanes, that are easy to get to on a bike.	92	81	94
There is a high crime rate in my neighborhood.	19	13	19
There is so much traffic along the street I live on that it would make it difficult or unpleasant to bike.	28	28	20
There is so much traffic along nearby streets that it would make it difficult or unpleasant to bike.	61	53	43
The speed of traffic on most nearby streets is usually slow.	56	53	64
Most drivers exceed the posted speed limits in my neighborhood.	68	73	61
Streets in my neighborhood are poorly maintained.	25	26	18

NOTE: Boldface indicates a significant difference from the utilitarian cyclist group, $p < .05$, two-tailed.

^a $n = 141$.

^b $n = 153$.

^c $n = 217$.

TABLE 6 Social and Personal Factors Among Interested but Concerned Group

Factor and Concern	Interested but Concerned (% agreeing)		
	Noncyclist ^a	Recreational Cyclist ^b	Utilitarian Cyclist ^c
Social			
Most people who are important to me, for example, my family and friends, think I should bike more.	33	29	48
Most people who are important to me, for example, my family and friends, would support me in using a bike more.	90	89	94
People I live with ride a bike to get to places, such as errands, shopping, and work.	17	17	53
Many of my friends ride a bike to get to places, such as errands, shopping, and work.	43	37	79
Many of my coworkers ride a bike to get to work.	36	30	51
In general, I see people similar to me bicycling on city streets.	71	76	85
Personal			
I feel a personal obligation to bicycle instead of driving for everyday travel.	18	17	54
For me to ride a bike for daily travel from home would be easy.	19	17	55
I know where safe bike routes are in my neighborhood.	69	78	94
Many of the places I need to get to regularly are within biking distance of my home.	63	47	73
I don't have time to bike places instead of driving.	62	69	49
Biking for commuting or transportation requires me to wear different clothes than normal.	72	84	69
I don't like wearing a bike helmet.	31	33	36
I know how to ride a bike safely in traffic.	83	93	100
I would like to learn how to ride more safely in traffic.	63	57	61
There is secure bike parking at my work or school.	73	73	83
I would feel comfortable riding my bike when it is raining.	23	28	47
I would feel comfortable riding my bike in my neighborhood after dark.	48	51	67
Concerns			
If or when I ride a bike, I'm concerned about . . .			
Being hit by a motor vehicle.	82	84	84
Being hit by another bicyclist.	12	14	18
Being bitten by a dog.	23	23	14
Falling off my bike.	32	21	23
Being stranded away from home.	31	25	20
Having my bicycle stolen.	64	63	63

NOTE: Boldface indicates a significant difference from the utilitarian cyclist group, $p < .05$, two-tailed.

^a $n = 141$.

^b $n = 153$.

^c $n = 217$.

concerned that they might be hit by a motor vehicle, although not by another bicyclist.

CONCLUSIONS AND POLICY IMPLICATIONS

The labeling or categorizing of cyclists has occurred for more than a century for a variety of purposes. Geller's typology developed for the City of Portland, Oregon, aimed to conceptualize the pool of existing and potential cyclists (4). As the proliferation of this typology in bicycle plans nationwide has demonstrated, Geller's typology has filled a need. Through a random survey of adults in the Portland region, the research reported in this paper concluded that, overall, the typology worked well to distinguish adults. Nearly all of the sampled population fit clearly into one of the four categories. The process

did use what could be considered arbitrary cutoff points to define respondent levels of comfort. To change these cutoff points, however, would simply affect the distribution of the population among the categories, and not whether or not someone fit into any one of the categories.

Geller's typology did appear useful to distinguish potential markets for cycling and to understand why some adults did not currently cycle for transportation. Some of the key findings and implications were as follows:

- Women were most likely to be in the no way, no how category or else to be noncyclists in the enthused and confident and the interested but concerned categories. The barriers that prevented these women from cycling for transportation needed to be better understood if cycling rates were to increase significantly. Other research

has indicated that common barriers include concerns about traffic, different attitudes toward bicycling, and complex travel patterns, including the transport of passengers (31).

- Older adults (older than 55 years of age) also were more likely to be in the no way, no how category or else to be noncyclists in the enthused and confident and interested but concerned categories. The large share in the no way, no how category reflected a physical inability to ride a bicycle among these respondents. Nontraditional bicycle technologies, including electric-assist bicycles (e-bikes) and adult tricycles, might be introduced to overcome this barrier for some older adults (34).

- A correlation was found between the experience of cycling to school as a child and the level of comfort with cycling as an adult. The enthused and confident adults were most likely to have cycled frequently to school as children, while most in the no way, no how group said that they never rode to school as children. Whether a person had cycled to school did not appear to affect whether an adult within one of the categories currently cycled for transportation or recreation. Because cycling frequency did vary by category, however, these findings did lend support to the hypothesis that increased cycling to school could have longer-lasting effects on overall rates of cycling.

- The interested but concerned adults did represent the largest potential market to increase cycling for transportation. Bicycle infrastructure that increased their physical separation from motor vehicles, such as cycle tracks, increased their reported level of comfort significantly. The provision of such infrastructure would seem a necessary condition to increase their levels of cycling for transportation.

- General concern about the amount of traffic and traffic speeds in neighborhoods, along with a lack of bike lanes and destinations nearby, appeared to prevent interested but concerned adults from bicycling. Traffic speed controls, traffic calming, and planning that promoted a mix of land uses could help overcome these barriers, in addition to bicycle-specific infrastructure.

- Social support for bicycling might influence whether an interested but concerned adult did cycle for transportation. Those that did not were less likely to live or work with people that bicycled for transportation or to see people that looked like themselves on bikes on city streets.

- Time constraints were an important barrier to cycling for transportation among the interested but concerned. To overcome this barrier would be challenging. Land use and street patterns that shortened travel distances between destinations could help, as well as more direct bicycle infrastructure. E-bikes might also be a solution for some adults.

- Self-reported knowledge of safe bicycling practices in traffic was high (more than 80% among the interested but concerned), although a majority of all subgroups did indicate an interest in learning more. These findings made it unclear whether or how much cycling education efforts would change levels of cycling. Interested but concerned adults that did not cycle felt considerably less comfortable with the idea of cycling in the rain or in the dark. Education might address such concerns.

The applicability of Geller's typology to other U.S. cities should be tested (4). It is likely that the distribution of any other region's population among the four types will differ. Until the study is replicated elsewhere, it is difficult to know if the magnitude of any differences is significant enough to affect the usefulness of this typology for planning purposes. The survey findings reported here indicated that the physical environment influenced the share of the population in each

category. A key question for further research to consider is whether and how much the physical environment for bicycling affects the two key components of the typology: comfort levels on different facilities and interest in cycling more. Other factors, such as the behavioral styles of drivers of motor vehicles (e.g., level of aggressiveness, speed) may also influence levels of comfort and interest.

Another focus of further research might be the subgroups of cyclists labeled as recreational cyclists, who were found here within all four categories of Geller's typology (4). They had ridden a bicycle in the past month but had not done so regularly for transportation. Very little research exists to examine the theory that people that cycle for recreation may transition more easily to cycling for transportation than people that do not bicycle at all. To test this theory, the use of longitudinal research, which perhaps involved an intervention, might be useful.

Finally, the aim of the study reported in this paper was to examine Geller's typology (4), which is used increasingly in other cities. It would perhaps be equally enlightening to develop a new typology altogether. A clear purpose and intended use are key to the development of an internally consistent and useful typology. As discussed above, one challenge is to separate actual bicycling behavior from levels of comfort and interest. A typology intended to help plan for a future with higher rates of cycling needs to apply to all adults, whether they bicycle or not.

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